Sahaj Bhadja

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Summary

Solutions Engineer with 2+ years of experience delivering **secure platform integrations**, **customer-facing workflows**, API-driven solutions, technical validation, and **cybersecurity risk applications** across production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable internal software workflows and technical operations.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production customer platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **maintainable service integrations**, indexing, caching, pagination, and filtered payloads while debugging workflows across Windows, Linux, and macOS production environments reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Increased lead conversions by 15% through **customer and lead-management experiences**, integrating role-based access, WebSocket updates, notifications, Azure SQL, and modular services across eight operational product domains reliably.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs through a **reproducible detection pipeline**, comparing Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models for phishing and malware identification.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **validated risk features** that exposed phishing and malware patterns reliably at scale.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash, C++
Security & Risk: Cybersecurity, User Authentication, Role-Based Access Control, JWT, Malware Detection, Phishing Detection
Solutions & Integrations: REST APIs, WebSockets, JSON, Dashboard Development, Workflow Automation, Customer Management
Applications & Data: React, Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Data Modeling, Query Optimization
Cloud & Delivery: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Cloud Storage
Quality & Reliability: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Fault Isolation, Production Reliability

Projects

Cyber Supply Chain Risk Operations - Design Concept
- Designed a **cyber supply-chain risk workflow** concept connecting vendor assessments, asset inventories, vulnerability feeds, and SBOM records through APIs with configurable scoring, dashboards, evidence tracking, and remediation workflows.

Blockchain Authentication Research | Live
- Designed a **distributed authentication architecture** spanning users, IoT devices, and role-based access scenarios, evaluating smart-contract verification, immutable identity records, and self-sovereign credentials against centralized identity dependencies reliably.

FinNexus | Live
- Built a **secure fintech platform** supporting 50K+ users and 1M+ transactions with TypeScript, WebSockets, advanced filtering, MongoDB-backed data access, token-based authorization, Kubernetes deployment, and synchronized interfaces reliably.